

Enterprise RAG Platform

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Validation

28 automated tests

Abstract

This brief documents a retrieval-augmented generation core built for enterprise knowledge use cases where accuracy, traceability, and tenant boundaries matter more than surface-level demo output.

Project Background

The operating problem was straightforward: teams keep policies, tickets, and wiki content in separate silos, but a useful assistant still has to retrieve the correct passage, cite it, and avoid leaking data across tenants.

Headline Result

Hybrid+Rerank: Recall@1 0.92, nDCG@10 0.96

Scope Overview

Streaming corpus generation and chunking with metadata. BM25, dense, hybrid retrieval, and learned reranking. Extractive synthesis with citation offsets. Evaluation harness, benchmarks, and visual scorecards.

Project Context

Production-grade retrieval-augmented generation - offline, deterministic, and evaluated.

Executive Summary

Background, objectives, scope, and project impact.

Project Background

The operating problem was straightforward: teams keep policies, tickets, and wiki content in separate silos, but a useful assistant still has to retrieve the correct passage, cite it, and avoid leaking data across tenants.

Executive Summary

This brief documents a retrieval-augmented generation core built for enterprise knowledge use cases where accuracy, traceability, and tenant boundaries matter more than surface-level demo output.

Objectives

- Raise first-result relevance without hiding weak baselines.
- Keep every answer grounded in recoverable source spans.
- Enforce tenant isolation inside retrieval, not after ranking.

Scope

- Streaming corpus generation and chunking with metadata.
- BM25, dense, hybrid retrieval, and learned reranking.
- Extractive synthesis with citation offsets.
- Evaluation harness, benchmarks, and visual scorecards.

Impact

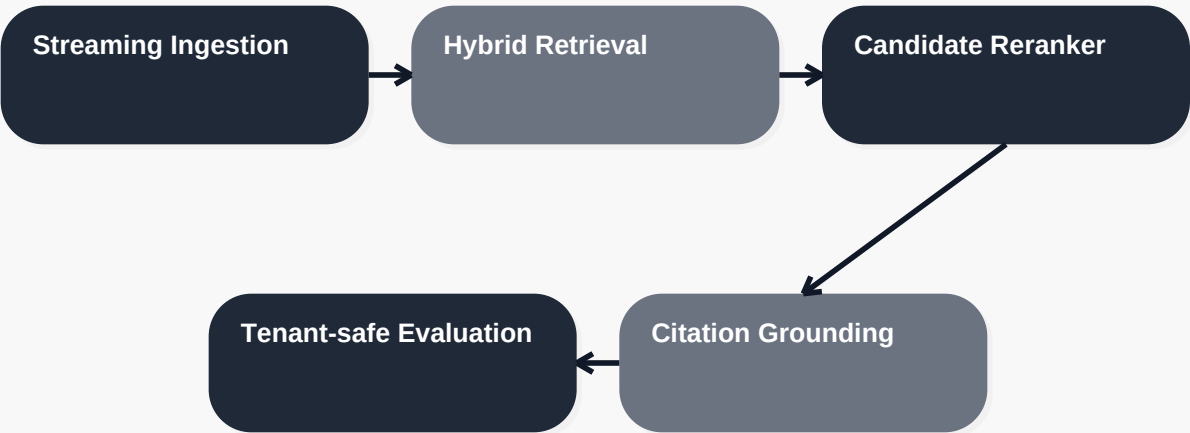
The final stack turned retrieval into a measurable engineering system. Hybrid retrieval with reranking improved top-rank quality while preserving grounded, citation-backed answers and deterministic offline reproducibility.

**Hybrid+Rerank: Recall@1 0.92,
nDCG@10 0.96**

Solution Architecture

Built around retrieval quality, citation traceability, and strict tenant isolation for enterprise knowledge systems.

Core system flow



Design Note 1

Hybrid fusion was kept score-scale agnostic so lexical and dense paths can coexist honestly.

Design Note 2

Citation grounding was treated as a product requirement, not a dashboard extra.

Design Note 3

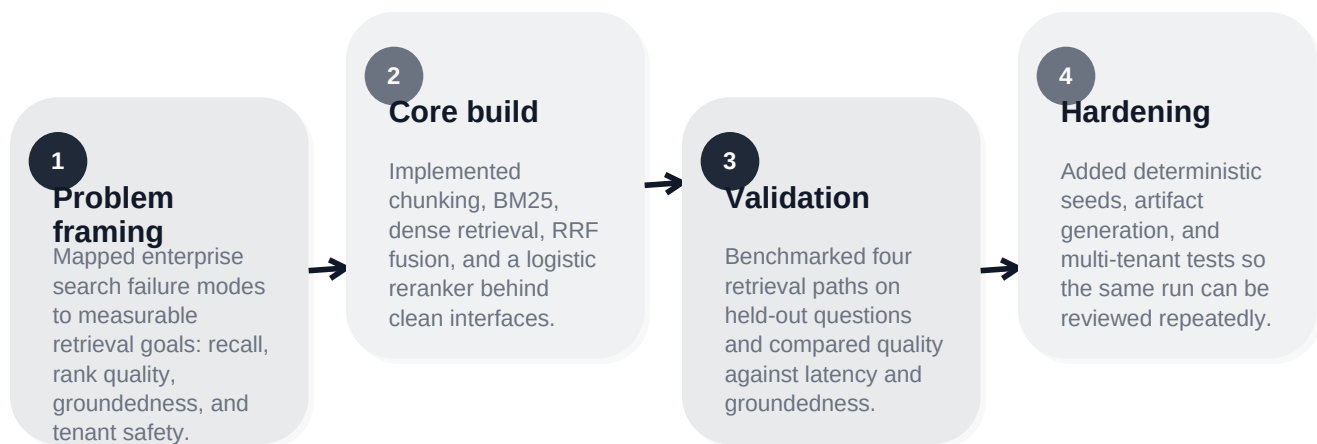
Isolation is applied inside the retriever to prevent ranking leakage and wasted top-k budget.

Implementation Process

Delivery phases, engineering approach, and quality checkpoints.

Delivery Narrative

Production-grade retrieval-augmented generation - offline, deterministic, and evaluated.



Quality Checkpoints

- 28 automated tests validate core behavior and edge cases.
- Benchmark files and screenshots are generated from committed project artifacts.
- The run path stays deterministic so numbers can be reproduced and reviewed directly.

Engineering Principles

- Hybrid fusion was kept score-scale agnostic so lexical and dense paths can coexist honestly.
- Citation grounding was treated as a product requirement, not a dashboard extra.
- Isolation is applied inside the retriever to prevent ranking leakage and wasted top-k budget.

Evidence And Results

Test evidence, benchmark interpretation, and screenshot review.

28 passing tests

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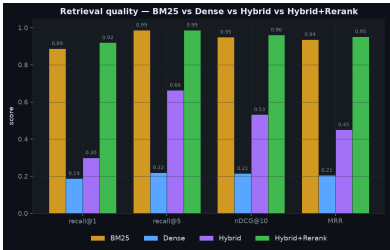
Hybrid+Rerank: Recall@1 0.92, nDCG@10 0.96

Benchmark Evidence

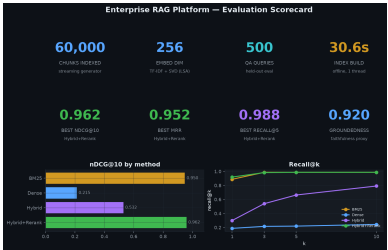
method	recall@1	recall@3	recall@5	recall@10
BM25	0.888	0.988	0.988	0.988
Dense	0.188	0.216	0.220	0.244
Hybrid	0.300	0.540	0.664	0.792
Hybrid+Rerank	0.920	0.984	0.988	0.988

Result Interpretation

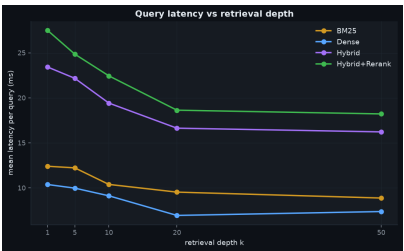
- The scorecards show Hybrid+Rerank leading both BM25 and dense-only retrieval on top-rank quality.
- Latency remains operationally plausible because the reranker only touches a bounded candidate set.
- Groundedness stays stable across retrieval modes because synthesis remains extractive and citation-aware.



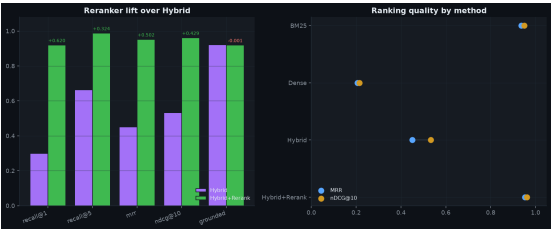
Retrieval quality - 4 methods compared



Evaluation scorecard



Query latency vs retrieval depth



Reranker lift & ranking quality

Risks And Next Steps

Operational considerations, mitigation choices, and forward path.

Risk Register

Tenant leakage

Mitigated by applying tenant masks before scoring rather than filtering results after ranking.

Dense recall weakness

Documented honestly and counter-balanced with hybrid fusion plus reranking.

Scale growth

Addressed by a sharded ANN path and streaming corpus generation strategy.

Next Steps

- Swap the deterministic embedder for a production encoder behind the same interface.
- Add online evaluation slices for business unit, tenant tier, and corpus freshness.
- Move from in-memory retrieval to shard-aware ANN infrastructure for larger corpora.

Closing Note

The project brief is intentionally written as delivered work rather than a transfer note. It documents the business problem, the engineering choices, the measured evidence, and the remaining operational path in a format suitable for portfolio review.

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